

SAURABH MISHRA

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TARGET ROLE: SOFTWARE ENGINEER (BACKEND / DISTRIBUTED SYSTEMS / PLATFORM ENGINEERING)

Backend Software Engineer with 4+ years of experience building large-scale, cloud-native backend systems in Java, Golang, and distributed architectures. Currently scale backend infrastructure for 10M+ users at Capslock Marketplaces, redesigning services around Kafka-driven async processing and Redis/Postgres caching. Previously shipped performance-critical Couchbase SDK enhancements sustaining ~10K ops/sec with zero failures. Strong in systems design, observability, and reliability engineering.

CORE SKILLS

Functional Skills: Backend System Design | Distributed Systems | Infrastructure Engineering | Event-Driven Architecture & Message Queues (Kafka, RabbitMQ) | Caching Strategy | High Availability & Fault Tolerance | Sharding & Replication | Load Balancing | Performance Optimization | Concurrency & Multithreading | REST & gRPC API Design | Monitoring & Observability (Prometheus, Grafana, OpenTelemetry) | CI/CD | Reliability Engineering | Production Debugging
Soft Skills: Problem-Solving, Documentation, Collaboration, Continuous Learning

TECHNICAL SKILLS

Languages: Java, Golang, .NET, Python, C++, Rust

Frameworks: Spring Boot, Gin, gRPC, REST, Reactive Java, RabbitMQ, Kafka

Software Design: OOD, LLD, HLD, Microservices, Distributed Systems, Event-Driven Architecture, Concurrency, System Design at Scale

Databases: Postgres, MongoDB, Redis, Couchbase, MySQL, Sharding, Replication

Infra & Tools: AWS, Azure, Docker, Kubernetes, Linux, CI/CD (GHA, Jenkins), Git, Maven, Prometheus, Grafana, OpenTelemetry

WORK EXPERIENCE

CAPSLLOCK MARKETPLACES, Bangalore — Software Development Engineer II (SDE II)

I Mar 2026 – Present

Own backend systems for the Voicelcub app, serving 10M+ users, focused on scalability, async processing, and real-time partner analytics.

- Integrated TeleCMI cloud telephony and OTPless passwordless authentication into the backend, enabling reliable call connectivity and frictionless partner onboarding for core voice workflows.
- Diagnosed a request-pileup issue caused by blocking synchronous calls and re-architected the affected services to process asynchronously via Kafka, removing a key latency bottleneck under peak load.
- Increased partition count on high-volume Kafka topics to improve consumer parallelism and throughput, reducing consumer lag during traffic spikes.
- Increased usage of the Redis caching layer in front of the primary Postgres database, cutting read latency and database load for high-traffic endpoints.
- Built a live analytics dashboard for partners to track real-time income and call volume, using Redis sorted sets to power live partner rankings.

COUCHBASE, Bangalore — Software Engineer II

I Aug 2022 – Mar 2026

Delivered customer-facing, performance-critical enhancements to Couchbase Java and .NET SDKs, improving scalability, observability, and reliability for distributed, cloud-native deployments.

- Promoted from Software Engineer to Software Engineer II for consistently delivering high-quality, high-impact contributions to Couchbase SDKs.
- Redesigned SDK network heuristics and connection resolution logic, resolving customer escalations under heavy load and sustaining ~10,000 ops/sec with zero failures.
- Designed and implemented Application Telemetry, enabling SDKs to publish runtime metrics for improved observability and diagnostics.
- Integrated Vector Search into Couchbase SDKs, enabling efficient similarity search for AI/ML-driven workloads.
- Developed Zone-Aware Replica functionality, improving data locality and reducing read latency in distributed deployments.
- Introduced Endpoint Events publishing, exposing DNS resolution and connectivity events for production debugging.
- Resolved critical SDK performance defects during cluster rebalancing by fixing memory leaks and node-targeting bugs.
- Migrated SDK build infrastructure from separate JVM/Mono versioning to a unified scheme, simplifying releases.
- Strengthened SDK test infrastructure with gRPC-based sanity and integration suites for high-concurrency validation.
- Built a custom proxy layer between SDK and server to intercept, manipulate, and replay requests for resilience testing.

ORANGE HEALTH, Bangalore — Software Engineer

I Jun 2022 – Aug 2022

Worked on backend systems for logistics optimization, focusing on scalability, routing intelligence, and real-time operations.

- Reduced total distance travelled by field medics by ~50% via Google Maps API integration and distance-based route optimization.
- Designed backend APIs for real-time logistics operations, improving visibility and coordination across stakeholders.

INSTAASTRO, Noida — Software Engineer Intern

I Feb 2022 – Jun 2022

- Designed and developed robust backend APIs supporting core features for web and mobile applications.
- Collaborated with frontend engineers and product managers to translate business requirements into scalable backend solutions.

PROJECTS

Sentinel Engine — Distributed Workflow Orchestration Engine

Java 21, Spring Boot 3, PostgreSQL, gRPC, Protobuf, Docker, OpenTelemetry, Prometheus, Grafana, GitHub Actions, Testcontainers

- Architected a Temporal-style durable workflow engine executing multi-step task DAGs with effectively-once semantics, surviving worker and network failures.
- Built a Postgres-backed durable task queue using SELECT FOR UPDATE SKIP LOCKED, validated zero duplicate claims across 50 concurrent workers on 100K+ tasks.
- Implemented DAG cycle detection and dependency-aware scheduling, sustaining 1,000+ task transitions/sec at p99 dispatch latency under 500ms.
- Engineered lease-based failure recovery with fencing tokens for idempotent re-execution, plus distributed tracing (OpenTelemetry) and Grafana dashboards; full unit/integration/chaos test coverage in CI/CD.

Distributed Key-Value Store (Golang + Rust)

Golang, Rust, gRPC, Raft Consensus, RESP Protocol, LSM-Tree, Protobuf, Docker, Kubernetes

- Built a distributed, Redis-compatible key-value store from scratch, implementing the RESP protocol and a Raft-based 5-node cluster with configurable linearizable/eventual consistency and automatic failover.
- Designed a Rust storage engine (write-ahead log (WAL), LSM-tree, background compaction) exposed to the Golang coordination layer via gRPC to avoid GC-induced write-path latency.
- Achieved 8,000+ writes/sec sustained throughput and sub-2s automatic leader failover under chaos testing; implemented horizontal sharding and a custom benchmarking harness.

EDUCATION

M.Sc., Artificial Intelligence & Machine Learning, BITS Pilani (Part-Time, In Progress)

I 2026 – 2028 (Expected)

B.Tech., Information Technology, ABES Engineering College, Ghaziabad

I 2018 – 2022

ACHIEVEMENTS

- Ranked 663rd globally among 25,000+ participants in a Codeforces contest.
- Finalist at TechGig Code Gladiators 2021; secured 1st place in Turing Test 6 coding competition (300+ teams).
- Solved 2000+ algorithm and data structure problems across competitive programming platforms.